

# Package ‘MIRT4FC’

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**Type** Package

**Title** Fit FC-Model by iStEM Alogrithm

**Version** 1.1.0

**Author** PeiYi Xu [aut, cre], YuKun Tu [aut, cre], ChanJin Zheng [aut]

**Maintainer** PeiYi Xu <xupy.ecnu@foxmail.com>, YuKun Tu <tuyukun1@outlook.com>

**Description** The objective of MIRT4FC is to efficiently implement various forced-choice models using the istem algorithm. Currently, it includes Thurstone's Item Response Theory (TIRT, Brown et al., 2011) Model, Multi-Unidimensional Pairwise Preference Two Parameter Logistic Model (MUPP-2PLM, Morillo et al., 2016) and we plan to continue updating and adding new models in the future.  
In addition to item parameter estimation capabilities, our R package also offers the ability to estimate ability parameters using MAP, EAP, and MLE methods. It can generate simulated response matrices, calculate standard errors (SE) for both ability and item parameters, and include a set of empirical data.

**License** GPL (>= 3)

**Encoding** UTF-8

**Data** inst/data/MAP\_data.rda

**LazyData** true

**ByteCompile** true

**RoxygenNote** 7.2.3

**Imports** armspp, doParallel, foreach, coda, stats, utils,  
Matrix

**Depends** R (>= 3.5.0), parallel, methods, mvnfast, nloptr

**Suggests** roxygen2

**SystemRequirements** GNU make

**NeedsCompilation** yes

## Contents

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|----------|-------------------------------------------------------------------------|
| data.sim | <i>Simulated dataset for multivariate FC item response theory model</i> |
|----------|-------------------------------------------------------------------------|

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**Description**

Simulated dataset for multivariate FC item response theory model.

**Usage**

```
data.sim(
  item.par = item.par,
  theta = theta,
  BID = BID,
  blocksize = 3,
  res = "rank",
  model = "2PL"
)
```

**Arguments**

|           |                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| item.par  | A data frame; parameters for a and d.                                                                                            |
| theta     | A # of subjects * # of dimensions matrix; theta parameters.                                                                      |
| BID       | A # of statements * 3 matrix; item information, columns are "Block", "Item" and "Dimensions".                                    |
| blocksize | A number; block size of FC(2/3/4).                                                                                               |
| res       | A string; response format('pick'/'rank'/'mole'), pick-2(blocksize=2)/rank-2/mole-2 are equivalent, rank-3/mole-3 are equivalent. |

**Value**

A # of subjects \* # of block number matrix.

**Examples**

```
##### A simulation example based on the MUPP-2PL model#####
#####Set simulation information
library (MIRT4FC)
D <- 6 # Dimension
nitem.per.dim <- 10 # Iems number per dimension
nblock <- D * nitem.per.dim / 3 # Blocks number
set.seed(123456) # Set random seed
# Simulate block-item-demension correspondence table
BID <- data.frame (Block = rep (1:nblock,each=3),
  Item=rep (1:3, nblock),
  Dim=c(combn(D,3) [, sample(choose(D,3), nblock,replace = TRUE)]))
# Simulate item parameter truth value
item.par <- data.frame (a = seq_len (D * nitem.per.dim))
item.par <- within (item.par, {
  a <- runif (D*nitem.per.dim,0.7,3)
  b <- rnorm (D*nitem.per.dim)
  d <- a*b
})
```

```

}))
item.par$d <- c(t(aggregate(item.par$d, by=list(BID$Block), function(x)x-mean(x))[, -1]))
N <- 1000 # Sample number
v <- matrix(0.5, D, D) # Intertrait correlation
diag(v) <- 1
# Simulate latent trait parameter truth value
theta <- mvnfast::rmvn(N, seq(-1, 1, length.out = D), sigma = v)
##### Generate a simulated dataset
Y <- data.sim(item.par, theta, BID)
print(Y)

```

iStEM

*Improved Stochastic EM algorithm for solving FC model item parameter*

## Description

Improved Stochastic EM algorithm for solving Force Choice: Two-Parameter Logistic (FC: 2PL) model.

## Usage

```

iStEM(
  Y,
  BID,
  positive = rep(TRUE, nrow(BID)),
  blocksize = 3,
  res = "rank",
  M = 10,
  B = 20,
  model = "2PL",
  SE = "Louis",
  sigma = NULL,
  theta = NULL,
  fix.sigma = FALSE,
  burnin.maxitr = 40,
  maxitr = 500,
  eps1 = 1.5,
  eps2 = 0.4,
  frac1 = 0.2,
  frac2 = 0.5,
  cores = NULL,
  h = NULL,
  fixed.blocks = NULL,
  fixed.a = NULL,
  fixed.d = NULL
)

```

## Arguments

Y                      A # of subjects \* # of blocks matrix; item responses.

|               |                                                                                                                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BID           | A # of statements * 3 matrix; item information, columns are "Block", "Item" and "Dimensions".                                                                                                                                   |
| positive      | A logical vector; indicating whether each statement is positive directional or not.                                                                                                                                             |
| blocksize     | A number; block size of FC(2/3/4).                                                                                                                                                                                              |
| res           | A string; response format('pick'/'rank'/'mole'), pick-2(blocksize=2)/rank-2/mole-2 are equivalent, rank-3/mole-3 are equivalent.                                                                                                |
| M             | A number; # of batch.                                                                                                                                                                                                           |
| B             | A number; # of iterations in each batch.                                                                                                                                                                                        |
| sigma         | A # of dimensions * # of dimensions matrix; initial sigma parameters.                                                                                                                                                           |
| theta         | A # of subjects * # of dimensions matrix; initial theta parameters.                                                                                                                                                             |
| fix.sigma     | Logical; TRUE if sigma is estimated.                                                                                                                                                                                            |
| burnin.maxitr | A number; max burn-in allowed.                                                                                                                                                                                                  |
| maxitr        | A number; max iterations allowed.                                                                                                                                                                                               |
| eps1          | A number; stability criteria.                                                                                                                                                                                                   |
| eps2          | A number; convergence criterion.                                                                                                                                                                                                |
| frac1         | A number; cutoffs for calculating Geweke z.                                                                                                                                                                                     |
| frac2         | A number; cutoffs for calculating Geweke z.                                                                                                                                                                                     |
| cores         | A number; number of cores.                                                                                                                                                                                                      |
| h             | A number; The perturbation constant of the differential method (Suggested values are 1e-5).                                                                                                                                     |
| fixed.blocks  | A vector of block indices whose item parameters (a, d) are kept unchanged during estimation; only the parameters of the other blocks are updated. Applies to all blocksize (2/3/4) and response formats.                        |
| fixed.a       | A matrix (blocksize x length(fixed.blocks)) of initial a values used as the fixed values for the fixed blocks. If NULL, the internal initial values are used.                                                                   |
| fixed.d       | A matrix (blocksize x length(fixed.blocks)) of initial d values used as the fixed values for the fixed blocks (only rows 1:2 are used; the third row is recomputed as -(d1+d2)). If NULL, the internal initial values are used. |

### Value

A list of data; including estimated a, d, and sigma parameters, total batch number, final chain size, burn-in size, time.

### Examples

```
D <- 6
nitem.per.dim <- 10
nblock <- D * nitem.per.dim / 3
set.seed(123456)
item.par <- data.frame(a=seq_len(D*nitem.per.dim))
item.par <- within(item.par,{
  a <- runif(D*nitem.per.dim,0.7,3)
  b <- rnorm(D*nitem.per.dim)
  d <- a*b
})
BID <- data.frame(Block=rep(1:nblock,each=3),
  Item=rep(1:3,nblock),
```

```

Dim=c(combn(D,3)[,sample(choose(D,3),nblock,replace = TRUE)]))
item.par$d <- c(t(aggregate(item.par$d,by=list(BID$Block),function(x)x=mean(x))[,,-1]))
N <- 1000
v <- matrix(0.5,D,D)
diag(v) <- 1
theta <- mvnfast::rmvn(N,seq(-1,1,length.out = D),sigma = v)
Y <- data.sim(item.par,theta,BID,blocksize=3,res='rank')
x <- iStEM(Y,BID,maxitr = 100,blocksize=3,res='rank',fix.sigma = TRUE,cores=1)

# Fix blocks 1 and 2 and set their (a, d) to specified initial values
fixed.blocks <- c(1, 2)
fa <- matrix(c(1.5,1.2,0.8, 2.0,1.0,1.3), nrow=3, ncol=2)
fd <- matrix(c(-0.5,0.3,0.2, -0.1,-0.4,0.5), nrow=3, ncol=2)
x2 <- iStEM(Y, BID, maxitr = 30, blocksize=3, res='rank', fix.sigma = TRUE, cores=1,
            fixed.blocks=fixed.blocks, fixed.a=fa, fixed.d=fd)

```

MAP\_data

*A Triple Empirical Data for Dominance Model*

## Description

This data set contains a real data set of 1391 participants on 88 triplets. In each triplet, participants had to rank the three alternative items according to their preference.

## Usage

```
MAP_data
```

## Format

A large matrix of 1391 observations containing information on 6 variables. In each block, 3 items(e.g. A/B/C) measure 3 different traits. A>B>C=1; A>C>B=2; B>A>C=3; B>C>A=4; C>A>B=5; C>B>A=6

## Examples

```

library(MIRT4FC)
Y <- data("MAP_data")
BID <- data.frame( # Block-item-dimension correspondence table
  Block = rep(1:88,each=3),
  Item = rep(1:3,times=88),
  Dim = c(11, 1, 14, 15, 13, 18, 6, 2, 7, 1, 21, 24, 23, 9, 22, 5, 3, 8,
6, 18, 20, 9, 7, 5, 19, 1, 9, 14, 22, 10, 23, 21, 8, 3, 20, 10, 22,
3, 17, 4, 23, 24,
8, 4, 24, 21, 12, 10, 9, 11, 8, 17, 12, 2, 15, 11, 14, 24, 22,
15, 13, 16, 14, 22, 4, 14, 10, 7, 6, 14, 19, 6, 4, 13, 3, 14, 15, 2,
8, 3, 11, 18, 23, 20, 24, 15,
22, 3, 20, 1, 14, 21, 16, 18, 4, 13, 16, 18, 5, 1, 8, 23, 2,
24, 11, 19, 23, 15, 12, 11, 10, 20, 9, 21, 10, 4, 16, 7, 2, 3, 12, 16,
10, 6, 13, 16, 21, 16, 17, 20,
10, 19, 13, 7, 5, 15, 11, 23, 24, 8, 6, 11, 2, 19, 15, 17, 20,
18, 9, 7, 12, 5, 9, 7, 22, 17, 24, 16, 6, 17, 13, 23, 7, 15, 17, 5,
8, 19, 14, 18, 3, 12, 22, 4, 5,

```

```

21, 13, 1, 23, 9, 13, 11, 22, 9, 19, 21, 8, 21, 12, 6, 16, 1,
23, 9, 1, 19, 2, 6, 11, 18, 24, 10, 7, 18, 17, 5, 7, 3, 22, 4, 2,
8, 20, 17, 15, 8, 3, 24, 12, 10, 22,
2, 11, 23, 19, 19, 13, 17, 6, 20, 24, 9, 17, 5, 20, 12, 6, 19,
18, 16, 15, 21, 7, 5, 1, 18, 2, 4, 14, 1, 13, 12, 16, 12, 2, 20, 4,
5, 10, 4, 1, 21, 14, 3))
##### Item parameter estimation
fit <- iStEM (Y, BID, maxitr = 150, blocksize = 3, res = 'rank', fix.sigma = TRUE)
print(fit)

```

---

theta.est

---

*Provide the function of estimating latent traits.*


---

## Description

Estimating latent traits for multivariate FC item response theory model.

## Usage

```

theta.est(
  Y,
  a = NULL,
  d = NULL,
  t = NULL,
  BID,
  sigma = NULL,
  prior = TRUE,
  blocksize = 3,
  res = "rank",
  model = "2PL"
)

```

## Arguments

|           |                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| Y         | A # of subjects * # of blocks matrix; item responses.                                                                            |
| a         | A vector; length = # of statements, initial alpha parameters.                                                                    |
| d         | A vector; length = # of statements, initial beta parameters.                                                                     |
| BID       | A # of statements * 3 matrix; item information, columns are "Block", "Item" and "Dimensions".                                    |
| sigma     | A # of dimensions * # of dimensions matrix; initial sigma parameters.                                                            |
| blocksize | A number; block size of FC(2/3/4).                                                                                               |
| res       | A string; response format('pick'/'rank'/'mole'), pick-2(blocksize=2)/rank-2/mole-2 are equivalent, rank-3/mole-3 are equivalent. |
| model     | A string; FC model('2PL').                                                                                                       |

## Value

A # of subjects x # of block number matrix

**Examples**

```
library(MIRT4FC)
D <- 6
nitem.per.dim <- 10
nblock <- D * nitem.per.dim / 3
item.par <- data.frame(a=seq_len(D*nitem.per.dim))
item.par <- within(item.par,{
  a <- runif(D*nitem.per.dim,0.7,3)
  b <- rnorm(D*nitem.per.dim)
  d <- a*b
})
a=matrix(item.par$a,nrow=3,ncol = 20)
d=matrix(item.par$d,nrow=3,ncol = 20)
BID <- data.frame(Block=rep(1:nblock,each=3),
  Item=rep(1:3,nblock),
  Dim=c(combn(D,3)[,sample(choose(D,3),nblock,replace = TRUE)]))
item.par$d <- c(t(aggregate(item.par$d,by=list(BID$Block),function(x)x-mean(x))[,,-1]))
N <- 1000
v <- matrix(0,D,D)
diag(v) <- 1
eigen(v)$values
theta <- mvnfast::rmvn(N,rep(0,each = D),sigma = v)
Y <- data.sim(item.par,theta,BID,blocksize = 3,res = "rank")
theta_est <- theta.est(Y,a,d,BID=BID,sigma=v,prior=TRUE,blocksize=3,res='rank',model='2PL')
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